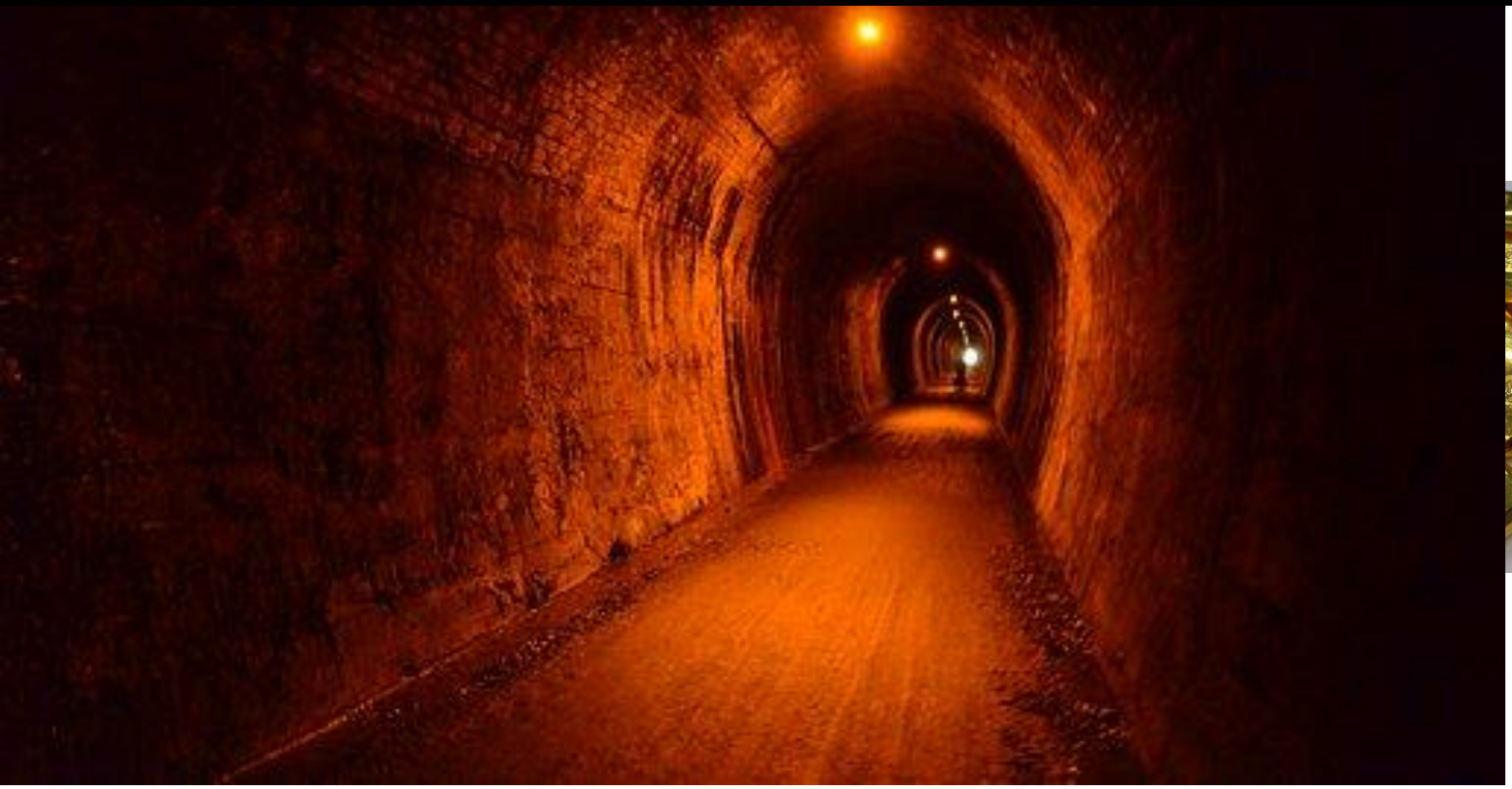
EART11200: Natural Scientist's Toolkit





Now use geometry to define the slope and direction of a seam of gold – the main problem for this week.

Structural Geology

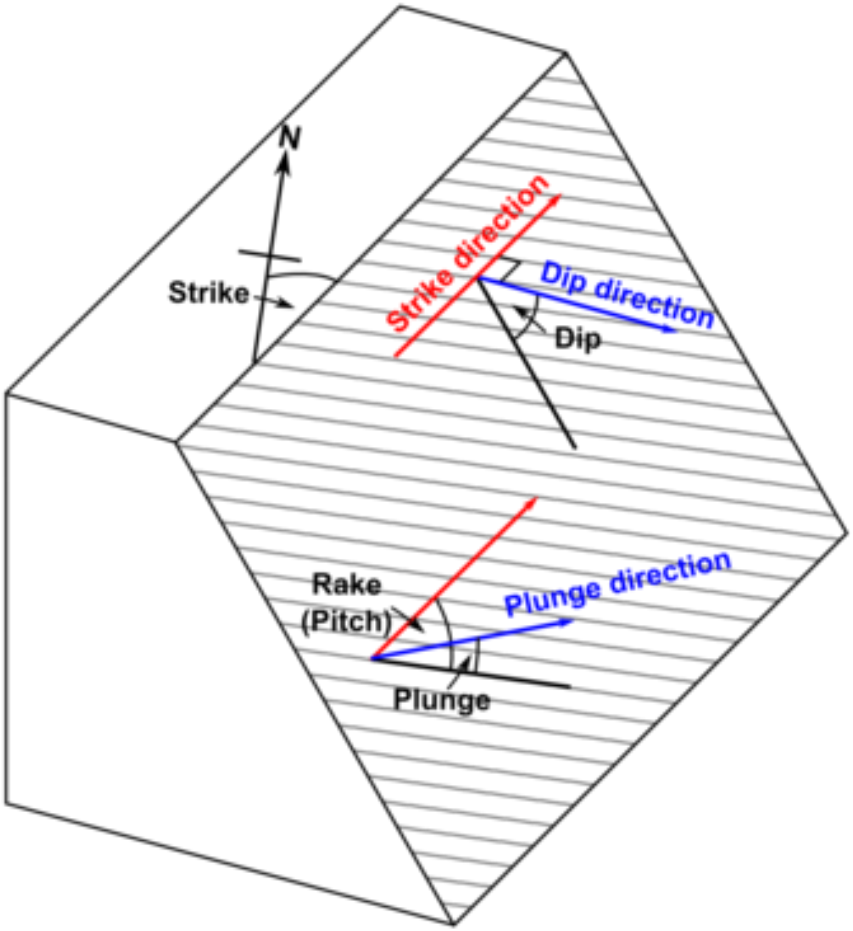
- Structural geology is the study of the three-dimensional distribution of rock units with respect to their deformational histories.

- Today we will look at a **seam of gold**, which is a plane of rock high in Au.

- Examples of planes:
  - Bedding, cleavage, joint surfaces, fault planes, fold axial planes, quarry faces

Geologists will measure some of these properties using a COMPASS-CLINOMETER, which you will have seen

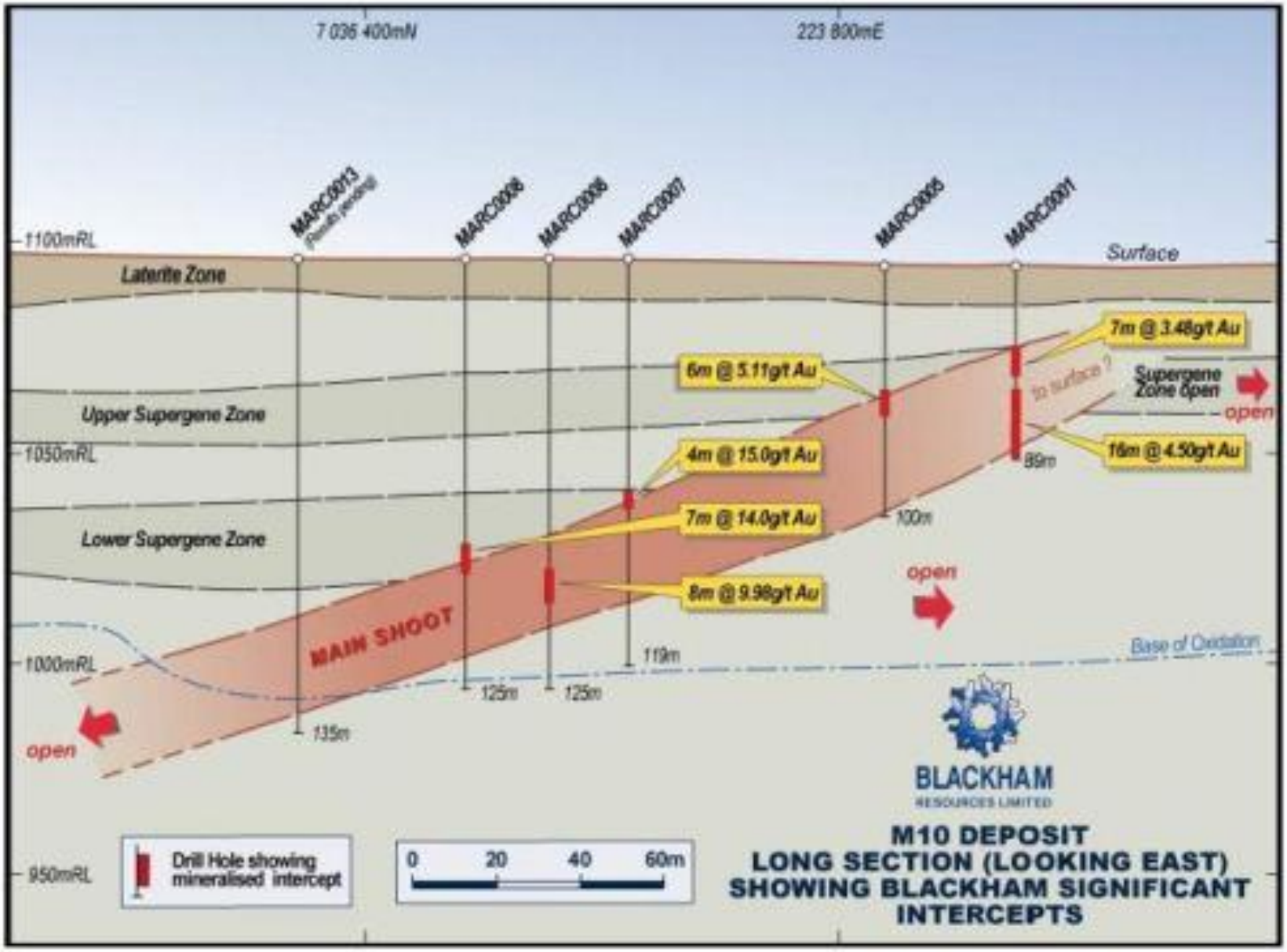
Geometries: Measurement conventions



- The inclination of a planar structure in geology is measured by strike and dip.
  - The **strike** is the line of **intersection between the planar feature and a horizontal plane**, taken according to the right hand convention,
  - the **dip** is the magnitude of the **inclination, below horizontal**, at **right angles** to strike.
  - For example; striking 25 degrees East of North, dipping 45 degrees Southeast, recorded as N25E,45SE.
- **Alternatively, dip and dip direction may be used as this is absolute**. Dip direction is measured in 360 degrees, generally clockwise from North. For example, a dip of 45 degrees towards 115 degrees from north, recorded as 45/115.

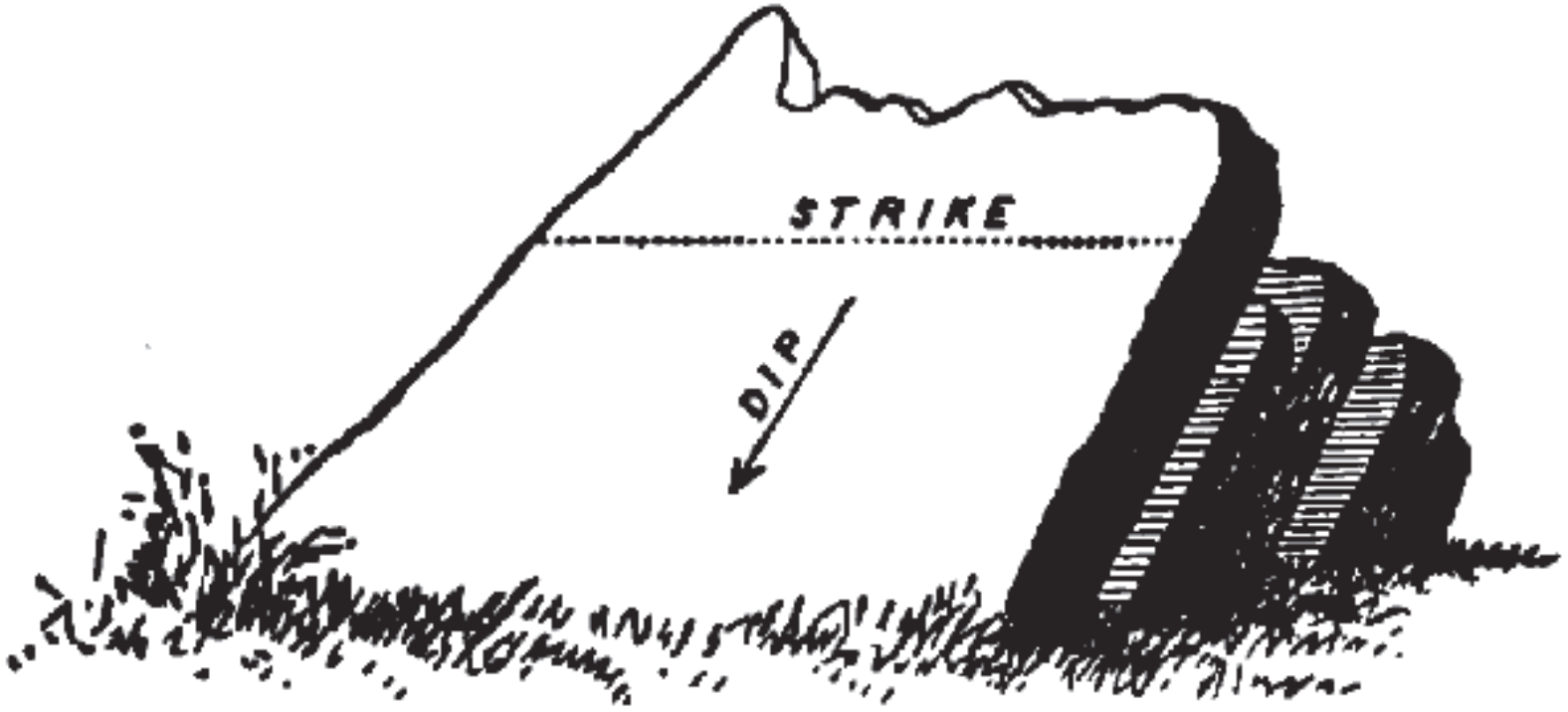
http://en.wikipedia.org/wiki/Structural_geology

Gold seam at Matilda, Australia (April 2012)



http://www.resourcesroadhouse.com.au/_blog/Resources_Roadhouse/post/Blackham_fires_up_new_drilling_program_at_Matilda/

Some times the seam will protrude out of the ground



1.4 A seam of gold

Problem 1.4 *A mineralised seam is visible at ground level but dips down into the surrounding rock. You are interested in mining this seam because it contains gold. However, you need to know whether it will be economic and therefore want to estimate the value of the gold it contains. To find out where it goes a hole has been drilled some distance away and the depth at which this seam reappears has been recorded.*

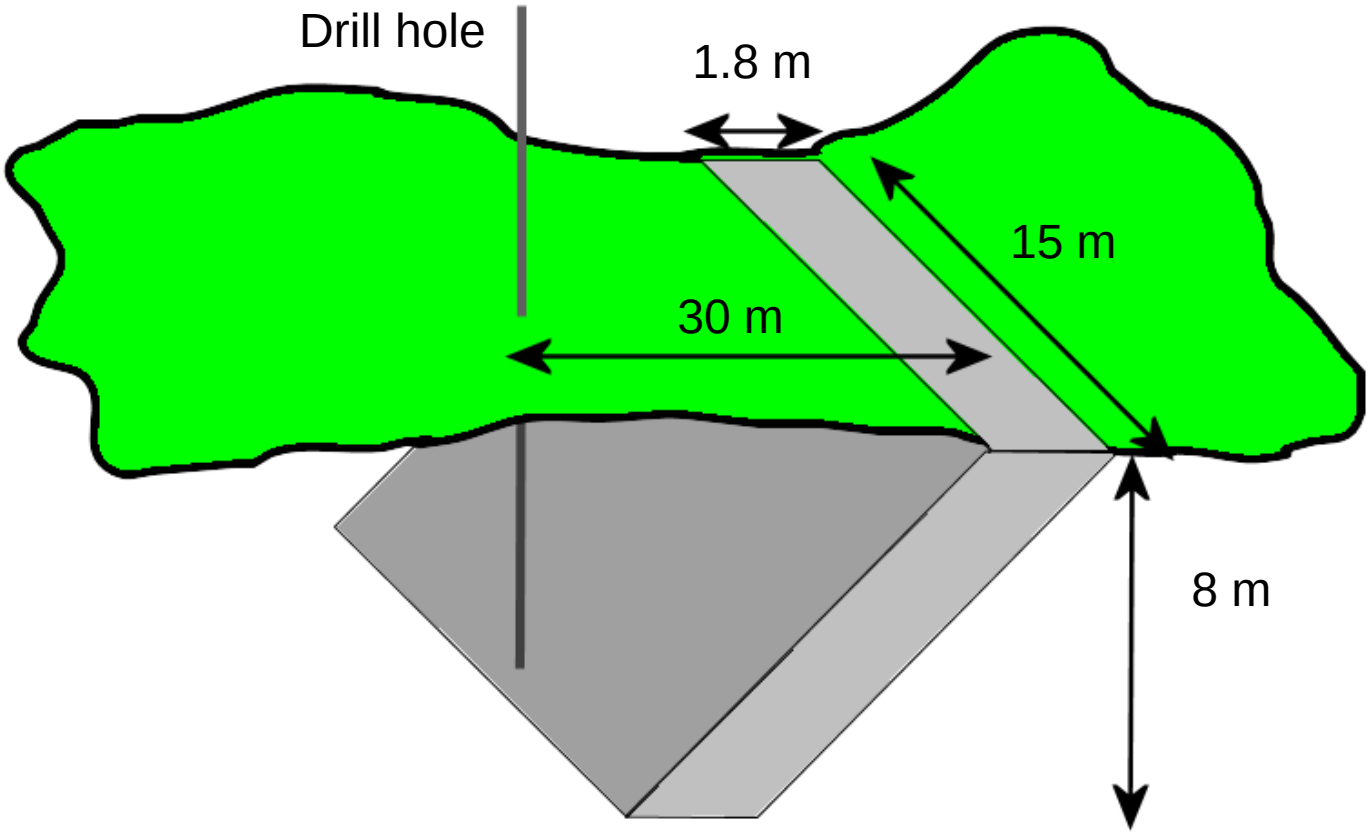
You can assume that the seam goes down at a constant angle and that you have followed it in the direction in which it dips most steeply.

Some information you might need:

- Horizontal distance from centre of seam to centre of borehole = 30m.
- Vertical distance from ground to centre of seam = 8m.
- Thickness of seam along ground = 1.8m
- Width perpendicular to dip = 15m.
- Density of ore 7.5 gcm^{-3}
- Concentration of gold in rock 3ppm by weight.
- Price of gold = £330 per troy ounce (NB. a troy ounce is 31.1g).

This is the main problem for this week. You do not have to complete it now, but have a think about it.

Lets model this as an idealised Gold seam



Those of you comfortable with Maths can solve this problem without the distance from centre of seam to bore hole (30 m).

Things for this week/practical

- Problems on trigonometry and Pythagoras' theorem (you can find additional help in Foundation Maths sections 4.1, 4.2, and 4.5)
- Main problem on seam of gold (and related problems if you can)
- Additional problem on Bragg's law
- If time try the sound pressure levels – not needed! You have to be familiar with logs for this and we have not covered logs yet.
- Blackboard Practice:
  - Seam of Gold
  - Angles, solutions of triangles and trig

- If you have not done A level maths, or are not familiar with the concept of logarithms, stop here.



- You need to do the main Seam of Gold problem and the Bragg's law problem. If you complete those, well done.
- If you want a challenge, continue for the theory behind the noise level problem.

1.4.4 Another related problem: noise levels

1.4.4.1 Theory

Pressure waves of a certain frequency are heard as noise. The amplitude of the pressure wave is referred to as the *sound pressure level*, p and the intensity is proportional to the square of the sound pressure level ($I \propto p^2$). Sound levels, L, are often quoted using the *decibel scale*, which is ten multiplied by the log to base 10 of the ratio of the square of sound pressure level to some reference pressure level, p_{rel}:

$$L = 10 \times \log_{10}\left(\frac{p_1^2 + p_2^2 + \cdots + p_n^2}{p_{rel}^2}\right)$$

here, $p_{ref} = 20 \times 10^{-6}$ Pa.

Often, the sound level of sound sources are quoted at some distance from the source. For example you may be able to find that the sound level of a car is 60 dB(A) at 10 meters. The intensity level of the sound source drops of as an inverse square law, which means that the pressure levels will drop off proportional to *inverse distance*, i.e. $p(x) = p_{lev} \times \frac{x_1}{x}$, where p_{lev} is the quoted sound pressure level at distance x_1 from the source.

Problem 1.5 *For example: the sound level of a car at 10 meters is 60 dB(A). What would be the sound level at 20 meters?*

We first calculate the sound pressure level

$$L = 10 \times \log 10\left(\frac{p^2}{p_{ref}^2}\right)$$

$\therefore$ dividing by 10 and raising both sides to the power 10 to eliminate the logarithm

$$10^{L/10} = \frac{p^2}{p_{ref}^2}$$

$\therefore$ take the square root of both sides and rearrange for p

$$p = p_{ref}\sqrt{10^{L/10}}$$

$\therefore$ substitute numbers

$$p = 20 \times 10^{-6}\sqrt{10^{60/10}}$$

$$p = 2 \times 10^{-2}\ Pa$$

we then need to scale the sound pressure level by inverse distance. i.e.

$$p(x) = p_{lev} \times \frac{x_1}{x}$$

$\therefore$ substitute the numbers from the problem

$$p(20) = 2 \times 10^{-2} \times \frac{10}{20}$$

$$p(20) = 1 \times 10^{-2}\ Pa$$

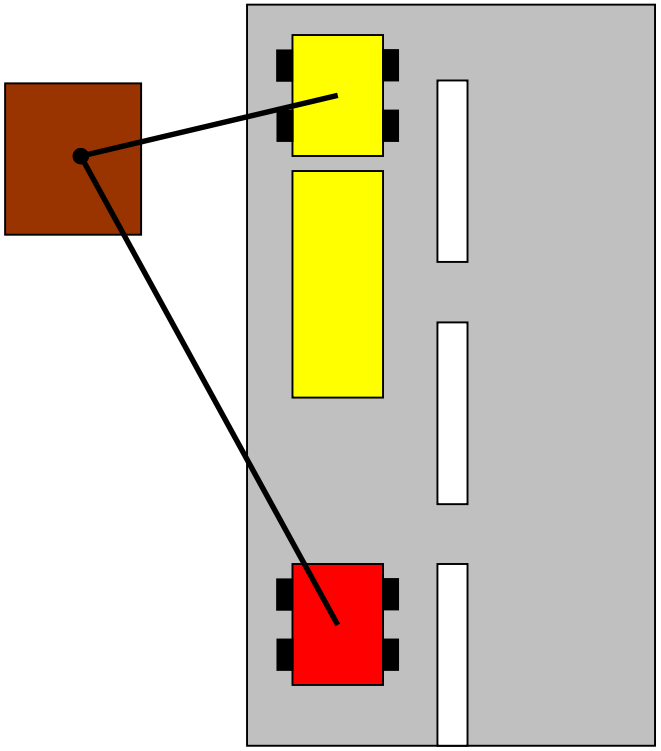
then to calculate the sound level that this pressure level corresponds to:

$$L = 10 \times \log_{10}\left(\left[\frac{1 \times 10^{-2}}{20 \times 10^{-6}}\right]^2\right)$$

$$L = \underline{53.98\ dB(A)}$$

P (rel/ref) comes from the previous slide and the definition of decibel.

Cars and HGVs noise levels



- Cars are 60 dB(A) at 10 m.
- HGVs are 84 dB(A) at 15 m.
- The house is 5 m off the road side.

1.4.4.2 Problem

1. A house is situated 300m from the start of a straight road and 5m from the edge of the road. There are two cars on the road at 200m and 700m, from the start of the road, that each have a sound level of 60 dBA measured at 10 m. There is also an HGV at 305m from the start of the road. The HGV has a sound level of 84 dB(A) at 15 m distance. What will be the sound level at the location of the house?